Fundamental Physics 1: Mechanics

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Definition of Power

- Machines such as wind turbines or engines do work by converting energy into useful forms.
- **Power** is the rate of doing work or using energy.
- Formula:

$$\text{Power} = \frac{\text{Work Done}}{\text{Time Taken}}$$

- Unit of power: **Watt (W)**
  - 1 Watt = 1 Joule per second

Examples of Power Usage

Activity/Device	**Power (W)**
Power for a small calculator	10^{-6}
Light power from a torch	10^{-3}
Loudspeaker output	10
Manual laborer working continuously	100
Water buffalo working continuously	10^{3}
Hair dryer	10^{3}
Motor car engine	10^{4}
Electric train	10^{6}
Electricity generating station output	10^{9}

Power and Energy Conversion

Power and Energy Conversion

- Power measures how quickly energy is converted from one form to another.
- For example:
  - A racing car and a family car may use the same amount of fuel.
  - However, the racing car is more powerful because it converts the energy of fuel faster.

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Power as a Scalar Quantity

- Power, like energy, is a scalar quantity.
- Common misconception:
  - Saying someone is "powerful" refers to strength, not physical power.
- For example, a large rock resting on the ground exerts a force but has no power as no work is done.

Formula for Power

- Work done by a force F moving an object over a distance x:

$$W = F \times x$$

- If the object moves at constant velocity v:

$$\text{Power} = \frac{W}{t} = F \times v$$

- Formula for power:

$$\text{Power} = \text{Force} \times \text{Velocity}$$

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$$\text{Average power} = \frac{\text{Work done}}{\text{Time interval}}$$

$$\Rightarrow \text{Power} = \vec{F}.\vec{v}$$

Worked example

A 1000 kg motorcar travels at a constant speed of 30 m/s along a road with a frictional force of 100 N. What is the power generated by the motorcar? ($g = 10$ m/s^2)

A 3000 W

B 4500 W

C 300 000 W

D 450 000 W

worked example

A small diesel engine uses a volume of 1.5×10^4 cm^3 of fuel per hour to produce a useful power output of 40kW. It may be assumed that 34 kJ of energy is transferred to the engine when it uses 1.0 cm^3 of fuel.
What is the rate of transfer from the engine of energy that is wasted?

A 102 kW

B 142 kW

C 182kW

D 470 kW

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Solution

A is correct

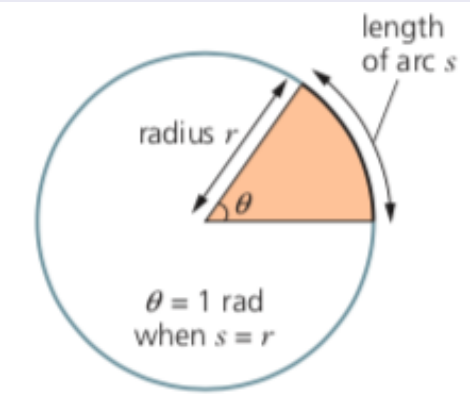
Worked example

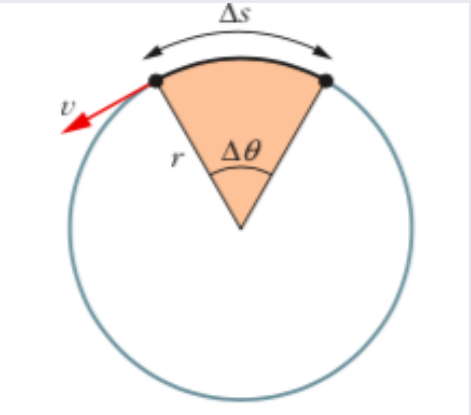
A car of weight 3000 N possesses an engine whose maximum power output is 160 kW. The maximum speed of this car on a level road is 35 m/s. Assuming that the resistive force (due to a combination of friction and air resistance) remains constant, what is the car's maximum speed on an incline of 1 in 20 (i.e., if θ is the angle of the incline with respect to the horizontal, then $\sin\theta = 1/20$)?

Kinematics of uniform circular motion

- Radian measure and angular displacement:
  One radian (rad) is defined as the angle subtended at the centre of a circle by an arc equal in length to the radius of the circle.

$$\theta = \frac{\text{length of arc}}{\text{radius of circle}} = \frac{s}{r}$$





- Angular speed: $\omega = \Delta\theta/\Delta t$
- Linear speed: $v = \omega.r$
- Period: $T = 2\pi/\omega$
- Frequency: $f = 1/T$

Worked example

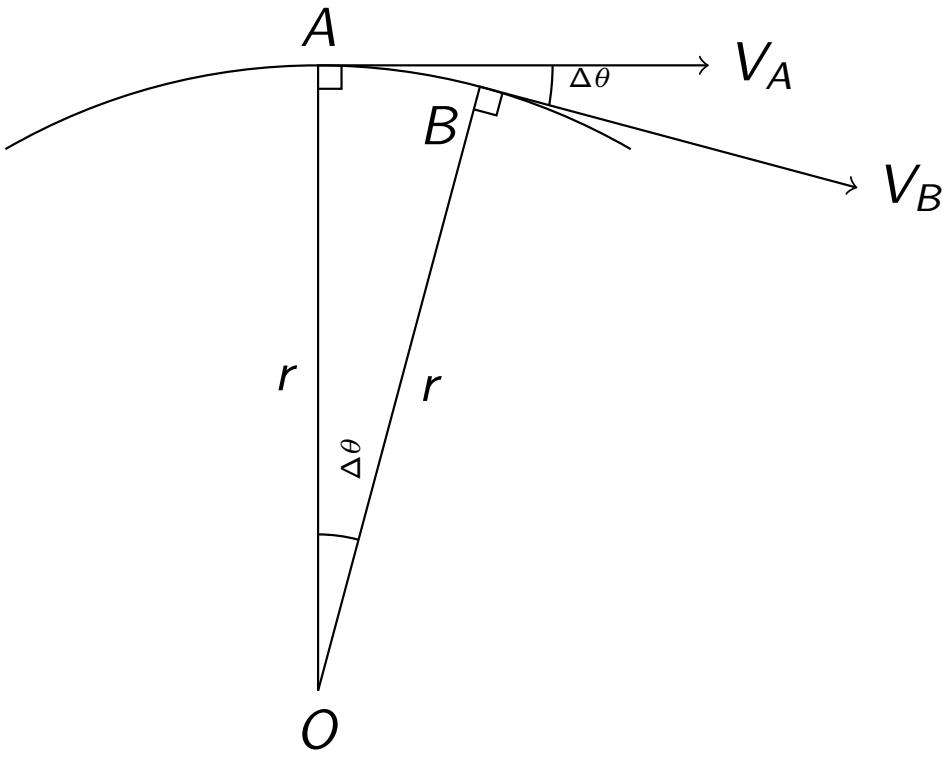
A phonograph record rotates at $33\frac{1}{3}$ revolutions per minute.

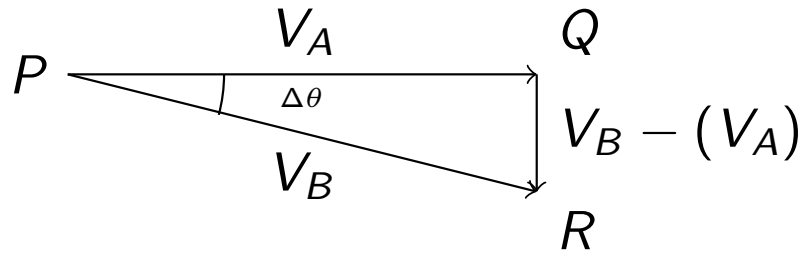
(a) Express the angular velocity of the record in radiant per second.

(b) What is the speed of a lady bird sitting on the record at a distance of

- 5.0 cm
- 10.0 cm from the center?

Chapter 4: Centripetal acceleration and centripetal force





- Centripetal acceleration: toward the centre of the circle.
- Magnitude: $a = \Delta v/\Delta t = \nu(\Delta\theta)/\Delta t = \nu\omega$
- Centripetal force:

$$F = m\nu\omega = mr\omega^2 = \frac{m\nu^2}{r}$$

Worked example

The drum of a spin dryer has a radius of 50cm and rotates at 720 revolutions per minute.

1. Show that the angular speed of the drum is about 75 rad s^{-1}
2. Calculate, for a point on the edge of the drum:
   - its linear speed
   - its acceleration towards the centre of the drum.

Worked example

A rope is tied to a bucket filled with water, and the bucket is swung in a vertical circle of radius 1.2 m. What must be the minimum speed of the bucket at the highest point of the circle if the water is to stay in the bucket throughout the motion?